

Affine-DDPNM: Revised Methodology

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This revision fixes the traction/load sign convention, the orientation of all tangential modes, the assumptions needed for local invertibility and global positive definiteness, and the interpretation of the Galerkin enrichment statement. The original files are not modified.

1 Model problem and weak-form convention

We consider steady incompressible Stokes flow in a pore domain $\Omega \subset \mathbb{R}^3$:

$$-\nu \Delta \mathbf{u} + \nabla p = \mathbf{0} \quad \text{in } \Omega, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0 \quad \text{in } \Omega, \quad (2)$$

$$\mathbf{u} = \mathbf{0} \quad \text{on } \Gamma_{\text{dir}}, \quad (3)$$

$$\boldsymbol{\sigma}(\mathbf{u}, p) \mathbf{n} = -p_b \mathbf{n} \quad \text{on } \Gamma_{\text{in}} \cup \Gamma_{\text{out}}, \quad (4)$$

where

$$\boldsymbol{\sigma}(\mathbf{u}, p) = -p \mathbf{I} + \nu \nabla \mathbf{u}.$$

The last condition is a prescribed pseudotraction, not a pressure Dirichlet condition. In particular, $\mathbf{n} \cdot \boldsymbol{\sigma} \mathbf{n} = -p_b$ does not imply $p = p_b$ pointwise because the normal viscous term is also present.

We use

$$b_i(\mathbf{v}, q) = - \int_{\Omega_i} q \nabla \cdot \mathbf{v} \, dx,$$

so that the local discrete system has the block form

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} U_i \\ P_i \end{bmatrix} = \begin{bmatrix} f_i \\ 0 \end{bmatrix}.$$

For a physical outward traction $\mathbf{t} = \boldsymbol{\sigma} \mathbf{n}$, the weak load is $\int_{\partial\Omega_i} \mathbf{t} \cdot \mathbf{v} \, dS$.

2 Partition and oriented interface frames

The domain is partitioned into non-overlapping Lipschitz subdomains $\{\Omega_i\}_{i=1}^N$. Each internal interface γ_k is shared by exactly two pores, denoted (i_k, j_k) , and is assumed planar in the theory. Choose once and for all a global orthonormal frame

$$\{\mathbf{n}_k, \mathbf{t}_{k,1}, \mathbf{t}_{k,2}\}$$

on γ_k , with $\mathbf{n}_k$ directed from i_k to j_k . Let $\sigma_{i,k} = +1$ for $i = i_k$ and $\sigma_{i,k} = -1$ for $i = j_k$. The local outward normal is

$$\mathbf{n}_{i,k} = \sigma_{i,k} \mathbf{n}_k.$$

To match the implementation in which the assembled normal load direction is $-\mathbf{n}_{i,k}$ and the assembled tangential load directions are $\sigma_{i,k}\mathbf{t}_{k,\nu}$, define the oriented inward-traction directions

$$\mathbf{d}_{i,k,n} = \mathbf{n}_{i,k}, \quad \mathbf{d}_{i,k,t_\nu} = -\sigma_{i,k}\mathbf{t}_{k,\nu}, \quad \nu = 1, 2. \quad (5)$$

Then

$$\mathbf{d}_{j,k,\beta} = -\mathbf{d}_{i,k,\beta} \quad (\beta = n, t_1, t_2).$$

The minus sign in the tangential definition is only an orientation choice; changing it together with the corresponding coefficient changes no traction space.

For the planar interface centroid $\mathbf{c}_k$, define dimensionless coordinates

$$s_k(\mathbf{x}) = \frac{(\mathbf{x} - \mathbf{c}_k) \cdot \mathbf{t}_{k,1}}{a_k}, \quad t_k(\mathbf{x}) = \frac{(\mathbf{x} - \mathbf{c}_k) \cdot \mathbf{t}_{k,2}}{b_k}, \quad (6)$$

with nonzero scales a_k, b_k . Let

$$\varphi_0 = 1, \quad \varphi_s = s_k, \quad \varphi_t = t_k, \quad \mathcal{M} = \{0, s, t\} \times \{n, t_1, t_2\}.$$

3 Affine traction Ansatz and code-compatible load sign

The unknown coefficient $\lambda_{k,\alpha,\beta}$ parameterizes the *inward* traction $-\sigma\mathbf{n}_{i,k}$:

$$-\sigma(\mathbf{u}_i, p_i)\mathbf{n}_{i,k} = \sum_{(\alpha,\beta) \in \mathcal{M}} \lambda_{k,\alpha,\beta} \varphi_\alpha \mathbf{d}_{i,k,\beta}. \quad (7)$$

Consequently the physical outward traction entering the weak form is

$$\sigma\mathbf{n}_{i,k} = - \sum_{(\alpha,\beta) \in \mathcal{M}} \lambda_{k,\alpha,\beta} \varphi_\alpha \mathbf{d}_{i,k,\beta}.$$

The primitive load vector is therefore

$$[\mathbf{f}_{i,k}^{(\alpha,\beta)}]_j = - \int_{\gamma_k} \varphi_\alpha \mathbf{d}_{i,k,\beta} \cdot \boldsymbol{\phi}_j^{(i)} dS. \quad (8)$$

This leading minus sign is essential. For the normal mode it gives the assembled direction $-\mathbf{n}_{i,k}$; for the tangential modes, using (5), it gives $+\sigma_{i,k}\mathbf{t}_{k,\nu}$, exactly the convention used in the supplied basis implementation.

At an inlet or outlet port only the constant normal mode is retained. The known coefficient is $\lambda^k = p_b$, so that the physical outward traction is $-p_b\mathbf{n}$.

4 Local finite-element response and compliance map

On pore Ω_i use an inf-sup stable velocity-pressure pair $(V_{i,h}, Q_{i,h})$. The rigorous finite-dimensional assumptions are:

Assumption 1. *For every pore i :*

1. A_i is symmetric positive definite. A positive-measure no-slip wall segment is a standard sufficient geometric condition for this coercivity.
2. B_i has full row rank in the pressure space actually used in the computation. This is a discrete inf-sup/rank assumption and is not implied by the wall condition alone.
3. The constant function 1 belongs to $Q_{i,h}$.

For each primitive mode solve

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} U_{i,k}^{(\alpha,\beta)} \\ X_{i,k}^{(\alpha,\beta)} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_{i,k}^{(\alpha,\beta)} \\ 0 \end{bmatrix}. \quad (9)$$

Let L_i collect all primitive load vectors, including one constant normal column for each known boundary port. Define

$$H_i = A_i^{-1} - A_i^{-1} B_i^\top (B_i A_i^{-1} B_i^\top)^{-1} B_i A_i^{-1}. \quad (10)$$

Then $U_i = H_i L_i \lambda_i$, and the local compliance matrix is

$$\mathbf{G}_i = L_i^\top H_i L_i. \quad (11)$$

It is symmetric positive semidefinite. For any local coefficient vector,

$$\lambda_i^\top \mathbf{G}_i \lambda_i = U_i^\top A_i U_i \geq 0. \quad (12)$$

For a local mode define the weighted velocity moment

$$M_{i,k}^{(\alpha,\beta)}(U_i) = \int_{\gamma_k} \varphi_\alpha U_{i,h} \cdot \mathbf{d}_{i,k,\beta} dS. \quad (13)$$

Because of the load convention,

$$[\mathbf{G}_i \lambda_i]_{k,\alpha,\beta} = -M_{i,k}^{(\alpha,\beta)}(U_i). \quad (14)$$

Thus $\mathbf{G}_i$ is the negative moment response associated with inward-traction coefficients. The sign has no effect on symmetry or energy positivity, but it must be retained in the interpretation and in the full-trace restriction.

5 Global Schur system

Let $\lambda \in \mathbb{R}^{9m}$ collect the shared unknown coefficient blocks on all internal interfaces. Since $\mathbf{d}_{j,k,\beta} = -\mathbf{d}_{i,k,\beta}$, using the same coefficient block on the two sides produces equal and opposite physical tractions. Continuity is imposed weakly through all nine moments:

$$M_{i,k}^{(\alpha,\beta)} + M_{j,k}^{(\alpha,\beta)} = 0. \quad (15)$$

Using (14), Boolean assembly gives

$$\mathbf{S} \lambda = F, \quad \mathbf{S} = \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u, \quad F = - \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \lambda^k. \quad (16)$$

Remark 1 (What is actually required for SPD). *Per-face linear independence of the nine scalar/vector functions is not sufficient for positive definiteness. The required local algebraic condition is*

$$\ker(H_i L_i) = \text{span}\{\mathbf{1}_{i,0n}\}, \quad (17)$$

where $\mathbf{1}_{i,0n}$ has value one in every constant-normal port slot and zero in all other slots. Equivalently, $\text{rank}(H_i L_i) = n_i - 1$. This condition excludes dependencies between different ports and dependencies created after the incompressible projection H_i . It must be proved for a specific discretization or checked numerically; it does not follow merely from independence on each face.

Under Assumption 1, condition (17), and the graph anchoring condition that every connected pore component contains at least one known-traction port, the companion mathematical document proves that $\mathbf{S}$ is SPD. The proof uses zero extension into the known-port slots; it does not apply the full local-kernel result directly to a shorter unknown vector.

Floating basins and graph anchoring are distinct issues. Merging a basin with no wall facet is intended to restore local coercivity of A_i . Ensuring that each connected pore component touches an inlet or outlet removes the remaining global constant-pressure gauge. One operation does not logically imply the other.

6 Algorithm

1. For each pore, assemble A_i , B_i , and verify the intended pressure rank convention.
2. For every internal port and every $(\alpha, \beta) \in \mathcal{M}$, assemble the signed load (8) and solve (9). For each known boundary port solve the single constant-normal response.
3. Form $\mathbf{G}_i = L_i^\top H_i L_i$. As a diagnostic, inspect the singular values of $H_i L_i$ or eigenvalues of $\mathbf{G}_i$ and verify that exactly one local null direction is present when (17) is assumed.
4. Assemble (16). Check symmetry and the smallest eigenvalue or a Cholesky factorization before treating the system as SPD.
5. Solve for λ and reconstruct each local field by linear superposition of primitive responses.
6. Verify local mass residuals, interface moment residuals, and the global boundary flux balance.

7 Relation to classic DD-PNM and scope of enrichment

Let $E_{P0} : \mathbb{R}^m \rightarrow \mathbb{R}^{9m}$ insert each classic coefficient into the constant-normal slot. Restricting both trial and test spaces gives

$$\mathbf{S}_{P0} = E_{P0}^\top \mathbf{S} E_{P0}, \quad F_{P0} = E_{P0}^\top F.$$

The full affine solution generally does not have the classic solution as its constant-normal sub-vector.

The valid enrichment statement is conditional and reference-dependent. If a well-defined full interface-traction Schur system $\widehat{S}\xi = \widehat{F}$ is introduced with the same local pressure spaces and sign conventions, then the affine system is its Galerkin restriction and the affine error is no larger than the classic error in the $\widehat{S}$ energy *seminorm*. This does not imply monotonic improvement of velocity L^2 , pressure L^2 , outlet flux, or broken H^1 errors.

No theorem in the present analysis establishes equality between that broken-pressure traction system and the monolithic continuous-pressure Taylor–Hood system. The supplied “Exact FE-trace Schur” computation is a primal Schur reduction of the monolithic system and therefore verifies its own algebraic elimination, not the missing equivalence with the broken-pressure traction formulation.

8 Computational cost

A pore with m_i^u internal ports and m_i^k known boundary ports has $n_i = 9m_i^u + m_i^k$ primitive columns. The offline response count therefore grows by approximately a factor of nine relative to a one-mode internal interface model. The global unknown count is $9m$. Sparse factorization cost depends on graph fill and cannot in general be inferred from the unknown-count factor alone.

References

- [1] S. Sun, Z. Wang, L. Zhang, and J. Zhao. A domain decomposition approach to pore-network modeling of porous media flow. *arXiv:2510.13429v2*, 2026.